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Research paper

Cooperative strategy based on a two-layer game model for inferior USVs to intercept a superior USV

Cong Chen [a], Xiao Liang [a,*], Zhao Zhang [a], Dianyong Liu [a], Wei Li [b]

[a] School of Naval Architecture and Ocean Engineering, Dalian Maritime University, Dalian, 116026, China
[b] School of Environmental Science and Engineering, Dalian Maritime University, Dalian, 116026, China

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ABSTRACT

Aiming at the problem of inferior USVs intercepting a superior USV, a cooperative interception strategy based on the two-layer game model is studied. Through the mathematical description of the USV's reachable set, the feasible interception point position is obtained. Two necessary conditions for completing regional defensive tasks under nonlinear motion conditions are proposed. By introducing the idea of a hierarchical task network, the complex regional defensive task is divided into different sub-tasks corresponding to the above necessary conditions. In order to better complete the switching between different sub-tasks, a two-layer game model is established. The selection of sub-tasks is modeled as an upper game, and the selection of basic maneuvers is modeled as a lower game. The Monte Carlo simulation results show that the cooperative interception strategy proposed in this paper has better performance under different conditions.

1. Introduction

1.1. Motivation and incitement

In recent years, Unmanned Surface Vehicles (USVs) have garnered increasing attention within the military domain (2021, Zhang Jie, Chen Zhiwen). This is attributed to their advantages, including cost-effectiveness, high mobility, and reduced risk (Ma et al., 2020). The evolving trajectory of advanced USVs focuses on facilitating demanding and intricate maritime operations, including post-disaster search and rescue, remote environmental monitoring and surveying, coastal patrols in conflict zones, and battlefield interception of suspicious targets (Zhang et al., 2020). Shortly, with the continuous advancement of sensor and robotics technology, USVs are poised to assume increasingly pivotal roles in the military domain (Zhou et al., 2022). Nevertheless, the combat capability of a single USV is limited, lacking support and guarantee, and cannot complete large-scale and complex tasks. To address this problem, the researchers propose that the cooperation among USVs can effectively compensate these limitations (Fu et al., 2019). Consequently, USVs are extensively deployed across a spectrum of combat missions (Wang et al., 2020).

As a prominent application, USVs cooperative interception focuses on researching autonomous cooperation of USVs to safeguard the protected area from intrusion. On the one hand, each defenders can employ various types of USVs with differences in maximum velocity and maneuverability. On the other hand, intruders often strive to possess superior performance to enhance the success probability of their intrusion tasks. Treating both sides of the game as homogeneous USVs makes it challenging to ensure the success probability of the interception task. Therefore, developing a cooperative strategy for the inferior USVs to intercept a superior USV is necessary.

1.2. Literature review

The cooperative interception problem of USVs is a multi-agent game confrontation problem, which has always been a research hotspot in multi-agent. With the wide application of unmanned equipment such as UAV, USV, and AUV, the application scope of this field has been further expanded. Various algorithms have been employed to tackle multi-agent game problems, including the guidance law, reinforcement learning, game theory, etc.

The utilization of the multi-agent game method based on guidance law is a widespread approach. Sun Zhiyuan et al.proposed a strategy of the multi-USV self-organizing cooperative pursuit of the escaper in open waters based on the Apollonian circle. The escape strategies of escapers in different hunting states are described (Sun et al., 2022). Taking the

* Corresponding author.
E-mail address: liangxiao@dlmu.edu.cn (X. Liang).

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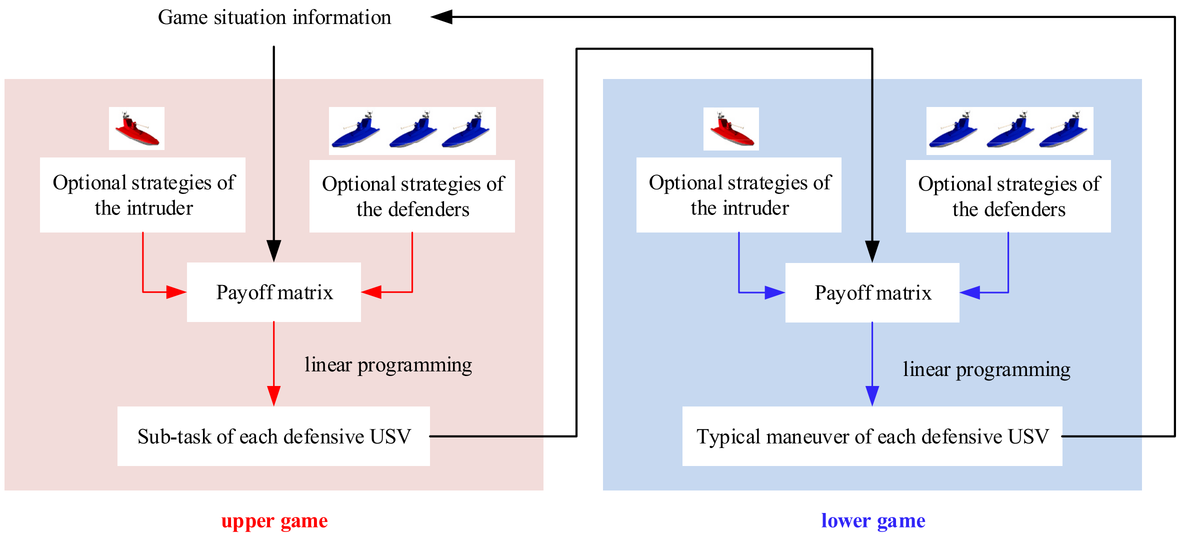


Fig. 1. Block diagram of the two-layer game model.

specific maneuverability and effective interception conditions of the gun-launched UAV in consideration, Li Jiguang et al. proposed a guidance law for the actual scene of the gun-launched UAV to intercept the 'low-slow-small 'target (Li et al., 2022).

The multi-agent game method based on reinforcement learning is also widely applicated in this field. Francesco Borra et al.proposed an adversarial reinforcement learning method for the pursuit-evasion problem of two tiny swimming agents in a low Reynolds number environment. The strategies agents adopt in different stages of adversarial learning are constantly evolving and updating. The results show that reinforcement learning can find more complex and effective game strategies and overcome the limitations of partial observability (Francesco et al., 2022). Aiming at the problem of confrontation games between USVs in a complex multi-obstacle environment, Qu Xiuqing et al. combined deep reinforcement learning with imitation learning to construct a USV pursuit decision model. At the same time, an adversarial-evolutionary game training method based on multiple random scenarios is proposed and combined with a curriculum learning method to train the pursuit and escape policy models. The simulation results show that the proposed method performs well in learning efficiency and game ability (Qu et al., 2023).

In the multi-agent game confrontation problem, the strategy selection between agents is mutual influence and restriction. Therefore, within multi-agent systems, game situation emerges due to the choice of strategies. Many scholars have also conducted extensive research on the application of game theory in multi-agent systems. Dave W. Oyler et al. proposed various methods to construct dominant regions and constructed a pursuit-evasion game model in an obstacle environment with asymmetric effects on players. An analysis of advantages provides a complete solution for the game, and then the optimal pursuit and escape strategies are obtained (Dave et al., 2016). Fang Xiao et al.studied the placement of multiple USVs from the game theory perspective. If the individual target of the USV may conflict with the global formation target of the group, the position game of multiple USVs is formulated. A game-based controller is designed for each USV, driving it to reach the equilibrium position that balances the formation target and the individual target (Fang et al., 2022). Xu Jiwei et al.proposed a multi-UAV confrontation model based on uncertain information, which can provide a powerful reference for strategy selection. The situation matrix is introduced to simulate the uncertainty of war information in establishing the game payoff function, and the interval decision model is established. The simulation results demonstrate that the proposed strategy can provide the optimal strategy for decision-making in uncertain information warfare (Xu et al., 2020).

The above research mainly focuses on one side of the balance game or the advantage side of the game, and there are relatively few studies on the disadvantage side of the game. For example, Yan Xinghui et al. proposed a novel cooperative strategy utilizing inferior missiles to intercept the highly maneuvering target. The reachability analysis is conducted using the control boundary of the target as a prior knowledge. The nonlinear kinematics is utilized in the reachability analysis, and the modified necessary condition of zero miss distance is obtained. In order to transform the quantitative advantage of missiles into operational advantages, the cooperative strategy aims to cover the reachable set of targets through the combination of missiles' reachable sets and establish favorable operational conditions (Yan et al., 2020). Building upon traditional reinforcement learning, Chi Pei et al. combined the grouping mechanism and proposed a decision-making method for UAV swarm confrontation based on multi-agent reinforcement learning. The results demonstrate that the approach can be extended to larger UAV swarms without increasing training time and performs well in attack-defense confrontation (Chi et al., 2023).

1.3. Contribution and paper organization

Following a comprehensive review of the existing research, we propose a multi-USV cooperative interception strategy based on the two-layer game model (TLG), as shown in Fig. 1. It is worth noting that the cooperative interception strategy proposed in this paper is also effective for invasive USVs with the same or lower motion performance. In a nutshell, the main contributions of this paper are as follows.

(1) We mathematically describe the reachable set of USVs in combination with motion performance and analyze the possibility of successful interception for each USV.
(2) We introduce two necessary conditions for the successful completion of regional defense tasks and employ a hierarchical task network to break down complex regional defense tasks into two simpler sub-tasks aligned with these necessary conditions.
(3) We establish a two-level game model, where sub-task selection is treated as an upper game, and the choice of typical maneuvers is represented as a lower game. This structure effectively translates quantitative advantages into game advantages.
(4) In order to fully illustrate the effectiveness and superiority of the cooperative interception strategy proposed in this paper, Monte Carlo simulation is used. Simulation results show that the

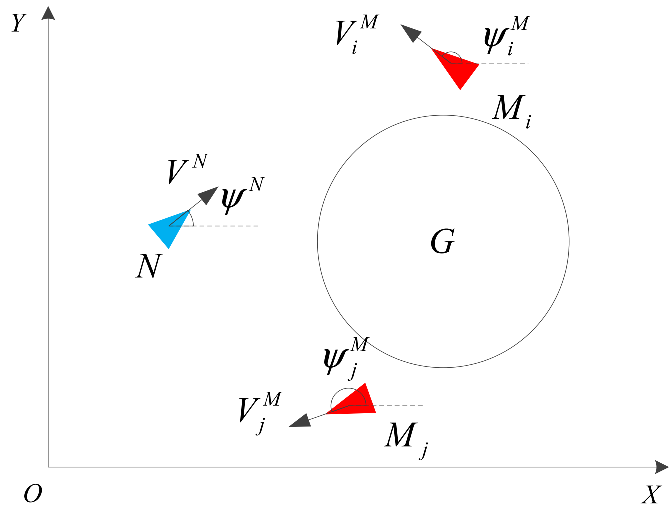


Fig. 2. Plane geometry of defensive task.

cooperative interception strategy proposed has better performance than the existing algorithms under various simulation conditions.

The organization of this paper is as follows: In Section 2, the mathematical description of the regional defensive problem and relevant assumptions is given. Section 3 discusses the reachable set of both sides of the game, introduces the concept of feasible interception points, and derives the equation for feasible interception points. A cooperative interception strategy based on the two-layer game model in proposed in Section 4. In Section 5, the performance of the proposed cooperative interception strategy is verified by Monte Carlo simulation. Lastly, Section 6 provides a summary and discussion of this paper.

2. Problem formulation

Considering the plane geometry between defensive USVs and the invasive USV, plane geometry of defensive task in the inertial Cartesian coordinates is illustrated in Fig. 2. $M = \{M_1, M_2, \ldots M_n\}$ represents a set of heterogeneous defensive USVs, and N represents an invasive USV. ψ_i^M represents the heading angle of the i th defensive USV, and ψ^N represents the heading angle of the invasive USV. G represents the protected area and it is also the target area of the invasive USV.

This study employs a widely-used representative USV model in inertial Cartesian coordinates. The second-order kinematic equation can be expressed as (Fan et al., 2020):

$$\begin{cases} \dot{x}_i = V_i \cos(\psi_i) \\ \dot{y}_i = V_i \sin(\psi_i) \\ \dot{\psi}_i = \omega_i \end{cases} \tag{1}$$

The relevant constraints are as follows (Su et al., 2017):

$$\begin{cases} V^N > V_i^M \\ \omega_{\max}^N > \omega_{i,\max}^M \end{cases}, i = 1, 2, \ldots, n \tag{2}$$

In the equation, x_i and y_i represent the position of the USV under the inertial Cartesian coordinates; ψ_i is the heading angle; V_i and ω_i represent velocity and angular velocity, respectively. equation (2) ensures that the invasive USV has advantages over any interception USV in maneuverability.

To simplify the theoretical analysis, the following assumptions are made.

(1) Each USV is considered a point mass.

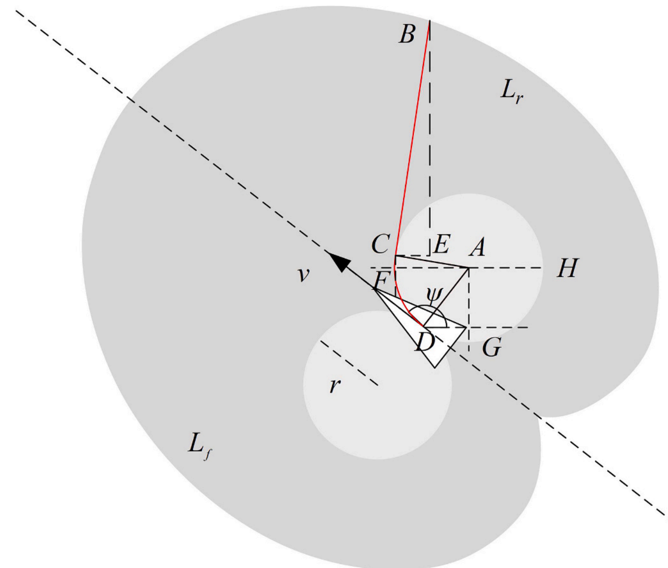


Fig. 3. Reachable set.

(2) The defenders are considered to be composed of heterogeneous USVs with different velocitys, angular velocities, and accelerations.
(3) The defenders can observe the situation information on the battlefield. However, there is an observation radius of the invasive USV, which can only observe the situation information within this range.
(4) The protected area's shape is considered circular.

3. Analysis of interception capability

In this study, the intruder has faster velocity and better maneuverability than the defenders. Consequently, in interception scenarios where the sole focus is on the current position of each USV, the intruder can swiftly evade the defenders' pursuit, ultimately reaching the protected area. To address this challenge, conducting a future-oriented analysis of potential states for both sides is essential, and determining the optimal interception strategy based on the current circumstances is essential.

3.1. Reachable set of USVs

可达集，即可达空间

In this paper, we define the reachable set $RS_{M_i}^{(t,t+\Delta t)}$ of the USV M_i between time t and time $t + \Delta t$ as the set of locations that all policy combinations in the policy set may reach (Zhou et al., 2018). Under the constraint of equation (1), the reachable set of USV as shown in Fig. 3.

The grey area in Fig. 3 represents the reachable range of the current USV, and $r = v/\omega$ represents the minimum turning radius. In a finite time Δt, the total length of the path of the USV at the maximum velocity remains unchanged. Therefore, the reachable set of the USV can be analyzed by the involute equations of the left and right circles formed by the minimum turning radius (Fumitaka et al., 2007), which are respectively recorded as L_f and L_r, and satisfy:

$$RS = L_f \cup L_r \tag{3}$$

Any point on the contour of L_f is located on the tangent of the minimum turning radius circle of the USV, and the point B coordinates can be expressed as:

$$(x_B, y_B) = \begin{pmatrix} x_C + |CB|\cdot\cos(\angle BCE), \\ y_C + |CB|\cdot\sin(\angle BCE) \end{pmatrix} = \begin{pmatrix} x_A + |AC|\cdot\cos(\angle CAH) + |CB|\cdot\cos(\angle BCE), \\ y_A + |AC|\cdot\sin(\angle CAH) + |CB|\cdot\sin(\angle BCE) \end{pmatrix} = \begin{pmatrix} x_D + |AD|\cdot\cos(\angle ADG) + |AC|\cdot\cos(\angle CAH) + |CB|\cdot\cos(\angle BCE), \\ y_D + |AD|\cdot\sin(\angle ADG) + |AC|\cdot\sin(\angle CAH) + |CB|\cdot\sin(\angle BCE) \end{pmatrix} \tag{4}$$

Suppose that the position of the USV at time t is (x_0, y_0), the heading is ψ_0, the velocity is v_0, the angular velocity is ω_0, and the central angle corresponding to the circular trajectory along the minimum turning radius is θ. It can be obtained that:

$$|AD| = |AC| = r = v_0/\omega_0 \tag{5}$$

$$|BC| = v\cdot\Delta t - \theta\cdot r \tag{6}$$

Therefore, the mathematical expression of L_r at time $t+\Delta t$ is:

$$L_r = \left\{ (x_r, y_r) \middle| \begin{array}{l} x_r = x_0 + (v_0/\omega_0)\cdot(\cos(\psi_0 - \pi/2) + \cos(\pi/2 - \theta + \psi_0)) + (v\cdot\Delta t - \theta\cdot v_0/\omega_0)\cdot\cos(\psi_0 - \theta), \\ y_r = y_0 + (v_0/\omega_0)\cdot(\sin(\psi_0 - \pi/2) + \sin(\pi/2 - \theta + \psi_0)) + (v\cdot\Delta t - \theta\cdot v_0/\omega_0)\cdot\sin(\psi_0 - \theta) \end{array} \right\} \tag{7}$$

Similarly, the mathematical expression of L_f at time $t+\Delta t$ is:

$$L_f = \left\{ (x_f, y_f) \middle| \begin{array}{l} x_f = x_0 - (v_0/\omega_0)\cdot(\cos(\psi_0 - \pi/2) + \cos(3\cdot\pi/2 - \theta - \psi_0)) + (v\cdot\Delta t - \theta\cdot v_0/\omega_0)\cdot\cos(\psi_0 + \theta), \\ y_f = y_0 - (v_0/\omega_0)\cdot(\sin(\psi_0 - \pi/2) - \sin(3\cdot\pi/2 - \theta - \psi_0)) + (v\cdot\Delta t - \theta\cdot v_0/\omega_0)\cdot\sin(\psi_0 + \theta) \end{array} \right\} \tag{8}$$

3.2. Feasible interception point

Definition 1. Possible interception point, which means that there is at least one strategy for the defender to meet the invader at this point in a limited time when the invader is sailing in the current state.

The possible interception points do not necessarily exist when the invader has a faster velocity. We first analyze whether there is a possible intercept point. The quintuple $S_t^i = \{x_i, y_i, v_i, \theta_i, \omega_i\}$ represents the state information of the USV i at time t, where each element represents the position, velocity, heading, and angular velocity of the USV. Therefore, the state information of the USV at time $t+\Delta t$ can be obtained by equation (9):

$$S_{t+\Delta t}^a = f(S_t^a, \Delta t) \tag{9}$$

where f represents the function of the USV's position changing with time. It is assumed that the invader sails at a constant velocity and heading. Substitute equation (1) for equation (9). The position of the invader at time $t+\Delta t$ can be expressed as:

$$\{(x_a, y_a) | x_a = x_{a0} + v_{a0}\cdot\cos(\theta_{a0})\cdot\Delta t, y_a = y_{a0} + v_{a0}\cdot\sin(\theta_{a0})\cdot\Delta t\} \tag{10}$$

equation (3) and equation (10) are solved simultaneously.

$$(x_d, y_d) = RS_M^{(t,t+\Delta t)} = (x_a, y_a) \tag{11}$$

Substitute equations (7), (8) and (10) for equation (11).

$$\begin{cases} x_d + r_d\cdot(\cos(\psi_d - \pi/2) + \cos(\pi/2 - \theta + \psi_d)) + (v_d\cdot\Delta t - \theta\cdot r_d)\cdot\cos(\psi_d - \theta) = x_a + v_a\cdot\cos(\psi_a)\cdot\Delta t \\ y_d + r_d\cdot(\sin(\psi_d - \pi/2) + \sin(\pi/2 - \theta + \psi_d)) + (v_d\cdot\Delta t - \theta\cdot r_d)\cdot\sin(\psi_d - \theta) = y_a + v_a\cdot\sin(\psi_a)\cdot\Delta t \\ \Delta t \geq 0 \\ \theta \geq 0 \end{cases} \tag{12}$$

$$\begin{cases} x_d - r_d\cdot(\cos(\psi_d - \pi/2) + \cos(3\cdot\pi/2 - \theta - \psi_d)) + (v_d\cdot\Delta t - \theta\cdot r_d)\cdot\cos(\psi_d + \theta) = x_a + v_a\cdot\cos(\psi_a)\cdot\Delta t \\ y_d - r_d\cdot(\sin(\psi_d - \pi/2) - \sin(3\cdot\pi/2 - \theta - \psi_d)) + (v_d\cdot\Delta t - \theta\cdot r_d)\cdot\sin(\psi_d + \theta) = y_a + v_a\cdot\sin(\psi_a)\cdot\Delta t \\ \Delta t \geq 0 \\ \theta \geq 0 \end{cases} \tag{13}$$

The possible intercept point exists if equation (11) and equation (12) have at least one real solution. Because the number of possible interception points is not unique in some cases, for a single defender, in the presence of multiple feasible interception points, we call the possible interception point with the least time the optimal interception point. The linear programming problem is as follows:

$$\text{argmin}(\nabla t)$$
$$s.t. \begin{cases} (x_d, y_d) = (x_a, y_a) \\ \Delta t \geq 0 \\ \theta \geq 0 \end{cases} \tag{14}$$

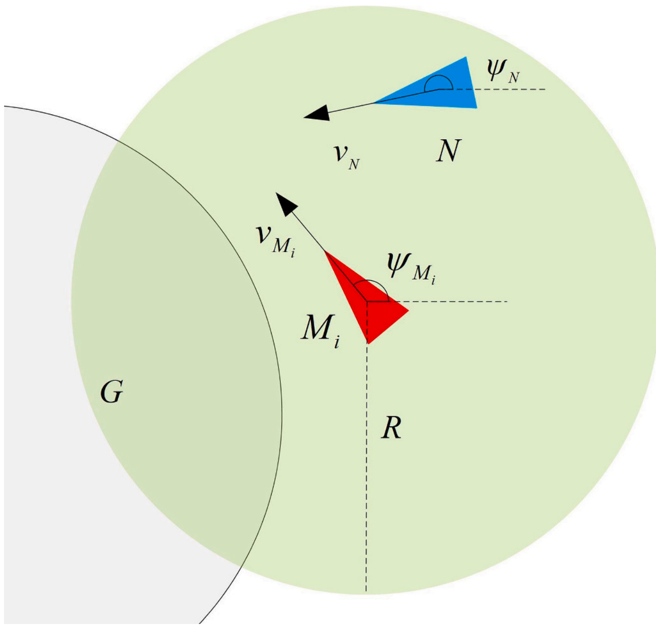


Fig. 4. Necessary condition 1.

4. Cooperative interception strategy

4.1. Necessary conditions for the successful completion of defensive tasks

完成拦截的两个必要条件，即激进策略和保守策略：

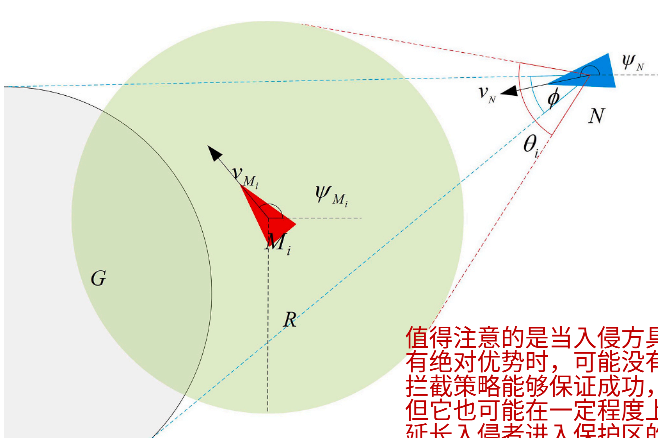
This section proposes two necessary conditions for completing defensive tasks under nonlinear kinematic conditions.

Necessary condition 1: Defenders complete the capture of the invader outside the protected area;

Necessary condition 2: Defenders prevent the invader from entering the protected area.

Definition 2. We call the angle of the two tangents passing through the intruder and tangent to the range of the defender's capture radius the safety range; the angle of the two tangents passing through the intruder and tangent to the protected area the danger range.

For the first necessary condition, the defender needs to reduce the distance from the invader as soon as possible and finally make the invader enter the capture range, as shown in Fig. 4. For the second necessary condition, the defender needs to make its safety range cover the dangerous range as much as possible so that the invader can not reach the protected area, as shown in Fig. 5. The defender M_i, when there is a possible interception point outside the protected area, it is



值得注意的是当入侵方具有绝对优势时，可能没有拦截策略能够保证成功，但它也可能在一定程度上延长入侵者进入保护区的时间。

Fig. 5. Necessary condition 2.

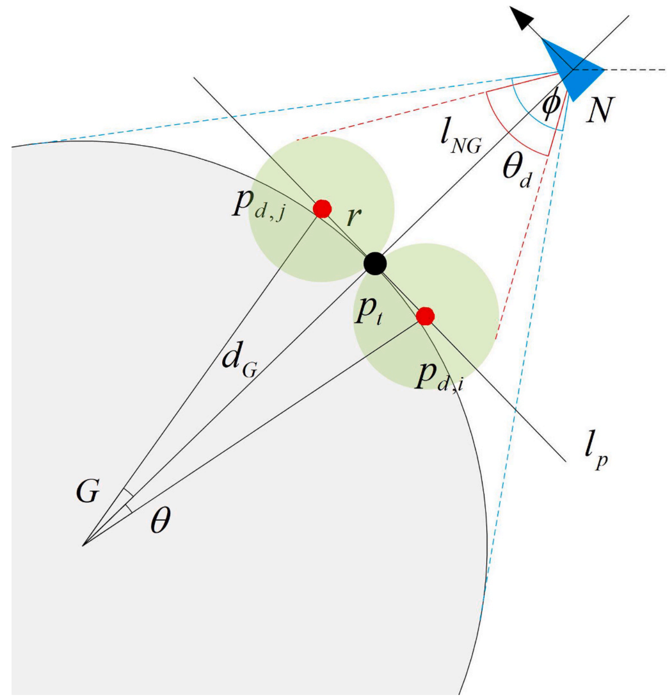


Fig. 6. Target points corresponding to the conservative task.

feasible for the defender M_i to complete the first necessary condition. The defender M_i is more inclined to complete the first necessary condition. Otherwise, the defender M_i is more inclined to complete the second necessary condition.

When the number of defenders is common, the interception strategy that can be adopted is more complicated. Determining the appropriate strategy for the current defender based solely on the presence of a potential interception point may not easily lead to the desired goal. Hence, a deeper analysis is required in such scenarios to make an informed choice regarding defender strategy. 采取两种任务分配策略的动机或原因：

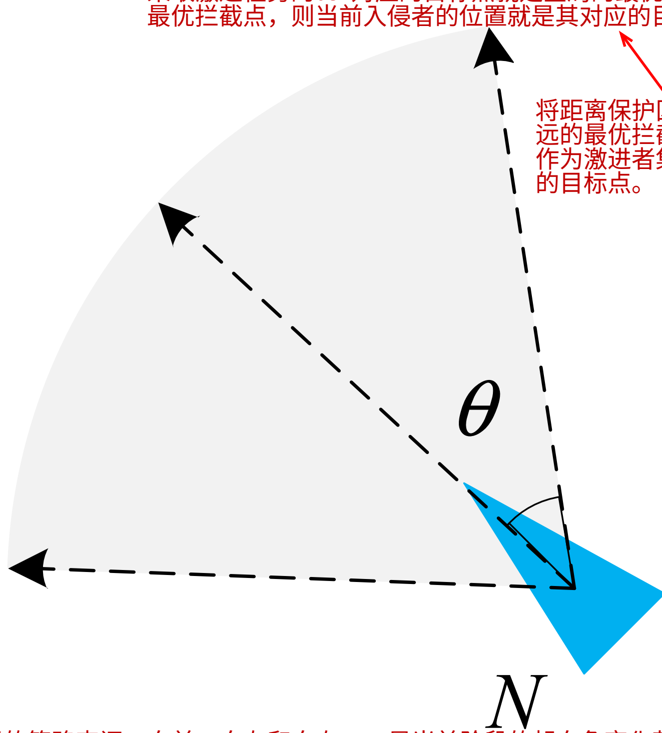
Assuming that the number of defenders with the optimal interception point at a particular time is common, we divide the feasible interception strategies into two types. The first is to select a part of the optimal interception points from each defender as the overall optimal interception point and select some defenders for each point, respectively. The remaining defenders tend to complete the necessary condition 2. The second is to select one of the optimal interception points of each defender as the overall optimal interception point and select some defenders to the point. The remaining defenders tend to complete the necessary condition 2. When the number of selected optimal interception points is not unique, the invader will be intercepted at the optimal interception point with the least time to take an evasive operation. Therefore, the defenders that go to the other optimal interception points cannot play their role. In the following study, we mainly focus on the second strategy.

As the distance between the invader and the protected area decreases, the level of threat increases. Therefore, we select the point with the farthest distance from the center of the protected area in the optimal interception point of each defender as the overall optimal interception point. However, a challenge persists. The forthcoming part will focus on the selection of appropriate strategies and implement the strategy to maximize the degree of defensive task completion.

It is worth mentioning that when the invader has an absolute performance advantage, no interception strategy can guarantee the completion of the defensive tasks. However, it can also prolong the time for the invader to enter the protected area to a certain extent.

这样分工的核心思想其实就是当防御方数量较多时，每个USV都有自己的最优拦截点，这样就会有多个最优拦截点，如果直接按照这个点各自追击，由于机动性能的劣势，很可能白忙活一场。本文选择一个全局最优拦截点（离受保护区域中心最远的那个），并派出一些USV作为激进方对其进行圈外拦截，剩下的防守在保护区外围进行拦截，这样进行有目的的分工才能更高效地对高性能目标进行拦截。

采取激进任务的USV对应的目标点就是全局的最优拦截点，若不存在全局最优拦截点，则当前入侵者的位置就是其对应的目标点。



将距离保护区最远的最优拦截点作为激进者集合的目标点。

入侵者的策略空间：向前、向左和向右，θ是当前阶段的朝向角变化范围

Fig. 7. Optional strategies for invader.

4.2. Multi-USV cooperative interception strategy based on the two-layer game model

Area defense is a time-series decision-making problem in complex environments (Sun and Yang, 2022). In the course of the mission, it becomes necessary to consider all potential states of each USV over a period. Suppose the basic actions such as velocity and heading are taken as the optional actions of each USV at each moment. This leads to an immense number of potential action combinations within a period, which will significantly increase the complexity of model construction and problem-solving, and it is challenging to meet the requirements of real-time confrontation (Brikaa et al., 2019). Therefore, more than just concentrating on the potential states of each USV at the next moment is needed.

海域区域防御是动态的连续决策问题，若直接让每艘 USV 在每个时刻选机动动作，会因组合太多导致模型复杂、决策变慢，满足不了海上攻防的实时性要求，因此必须做“分层决策”。

4.2.1. Two-layer game model

We introduce the concept of a hierarchical task network to decompose complex tasks into a series of more specific sub-tasks (Luan et al., 2021). A cooperative interception strategy is proposed based on the two-layer game (TLG). The upper game model takes a higher-level sub-task as the optional action of the participant; the lower game model takes the primary action of the USV as the optional action of the participant (Fang et al., 2020).

Both the upper game model and the lower game model can be modeled by a multivariate array $\langle P, S, A, R\rangle$ (Mkiramweni et al., 2019). Here, P represents the set of participants in the game, represents the set of defenders and invader. $S = \{s_0, (S_i)_{i\in P}, s_E\}$ denotes all possible state sets, where s_0 denotes the initial state, $(S_i)_{i\in P}$ denotes all possible intermediate state sets of participant i, and s_E denotes all possible final state sets. A and R represent the set of participants ' optional strategies and the set of payoffs, respectively, which will be explained later.

4.2.2. Optional strategies

In the upper game, we decompose the complex defensive task into the conservative task and radical task as the optional strategy of each defender. All possible strategy combinations of the defenders are optional strategies for the defenders (Liu et al., 2022). The two sub-task correspond to different target points. Among them, defenders taking the radical task aim to complete the necessary condition 1, and the corresponding target point is the overall optimal interception point of defenders. When the overall optimal interception point does not exist, the corresponding target point is the current position of the invader. The defender adopting the conservative task aims to complete the necessary condition 2, and the corresponding target points are shown in Fig. 6.

在上层博弈中，将防御任务分解为保守任务和激进任务。

In the figure, θ_d represents the expected safety range when the defenders adopt the current strategy; φ denotes the dangerous range at the current moment; l_{NG} is the connection between the current invader and the position of the protected area; p is the intersection of l_{NG} and the contour of the protected area; l_p is a tangent line of the protected area through the p; r is the capture radius of the defenders; d_G is the radius of the protected area.

The number of target points is the same as the number of defenders taking the conservative task. The angle between the i th target point and its adjacent target points is $2{\cdot}\theta$, where $\theta = \arctan(r/d_G)$, and satisfies equation (15):

$$z(p_t) = \sum_{i=1}^{n_d} z(p_{d,i}) \Big/ n_d \tag{15}$$

根据均匀部署在保护区的保守者USV的位置坐标计算保守者集合对应的目标点。

Where $z(p_t)$ denotes the current position of the point p_t; $z(p_{d,i})$ denotes the current position of the point $p_{d,i}$; n_d denotes the number of target points. For the invader, we use straight, left-maneuvering, and right-maneuvering as the optional actions, as shown in Fig. 7, where θ represents the maneuvering range, that is, the heading change range of the invader in a period.

上层博弈中入侵者的动作空间

In the lower game, nine typical maneuvers were used as optional strategies for both sides, which are uniform straight, uniform left turn, uniform right turn, deceleration straight, deceleration left turn, deceleration right turn, acceleration straight, acceleration left turn, acceleration right turn (Sun et al., 2023). The maximum acceleration and the maximum angular velocity realize the realization of each maneuver.

下层博弈中双方的动作空间

4.2.3. Payoffs

上下层的支付计算的都是防御方团队的效用而非个体效用！

$M = \{M_1, M_2, \ldots, M_n\}$ is the set of defenders; N is an invader. In the upper game, let $A = \{A_i\}_{i\in M}$ be a binary decision array. If the defender takes a radical task, then A_i is 1, otherwise A_i is 0. The construction of the upper game comprehensive payoff function is as follows (Guo et al., 2022): 防御者的上层决策变量是二元变量，1为激进，0为保守。

$$R_h = k_1{\cdot}r_1 + k_2{\cdot}r_2 + k_3{\cdot}r_3 \tag{16}$$

In the equation, k_1, k_2 and k_3 represent the specific gravity coefficient; r_1 represents the interception capability payoff, that is, the payoff obtained by defenders taking the radical task, as shown below:

激进者的拦截能力支付

$$r_1 = e^{-\sum_{i=0}^{n}(T_i - \min(T)){\cdot}A_i} \tag{17}$$

$T = \{T_1, \ldots, T_n\}$ is the set of the time taken by each defender to reach the current target point.

r_2 represents the synergy payoff, that is, the payoff obtained by the synergy effect between the defenders adopting the radical task, as shown below:

激进者之间的协同效应支付

$$r_2 = \sum_{i=1}^{n} \frac{(2{\cdot}d_c - d_i){\cdot}A_i}{2{\cdot}d_c} \tag{18}$$

In the equation, d_c represents the capture radius of each defender; d_i represents the distance between the i th defender and the invader when the first defender reaches the target point.

r_3 represents the payoff of responding to potential threats, that is, preventing the intruder from evading interception by its mobility and successfully invading. The payoff obtained is related to the maximum proportion of the safety range in the danger range, as shown below (Wang et al., 2022):

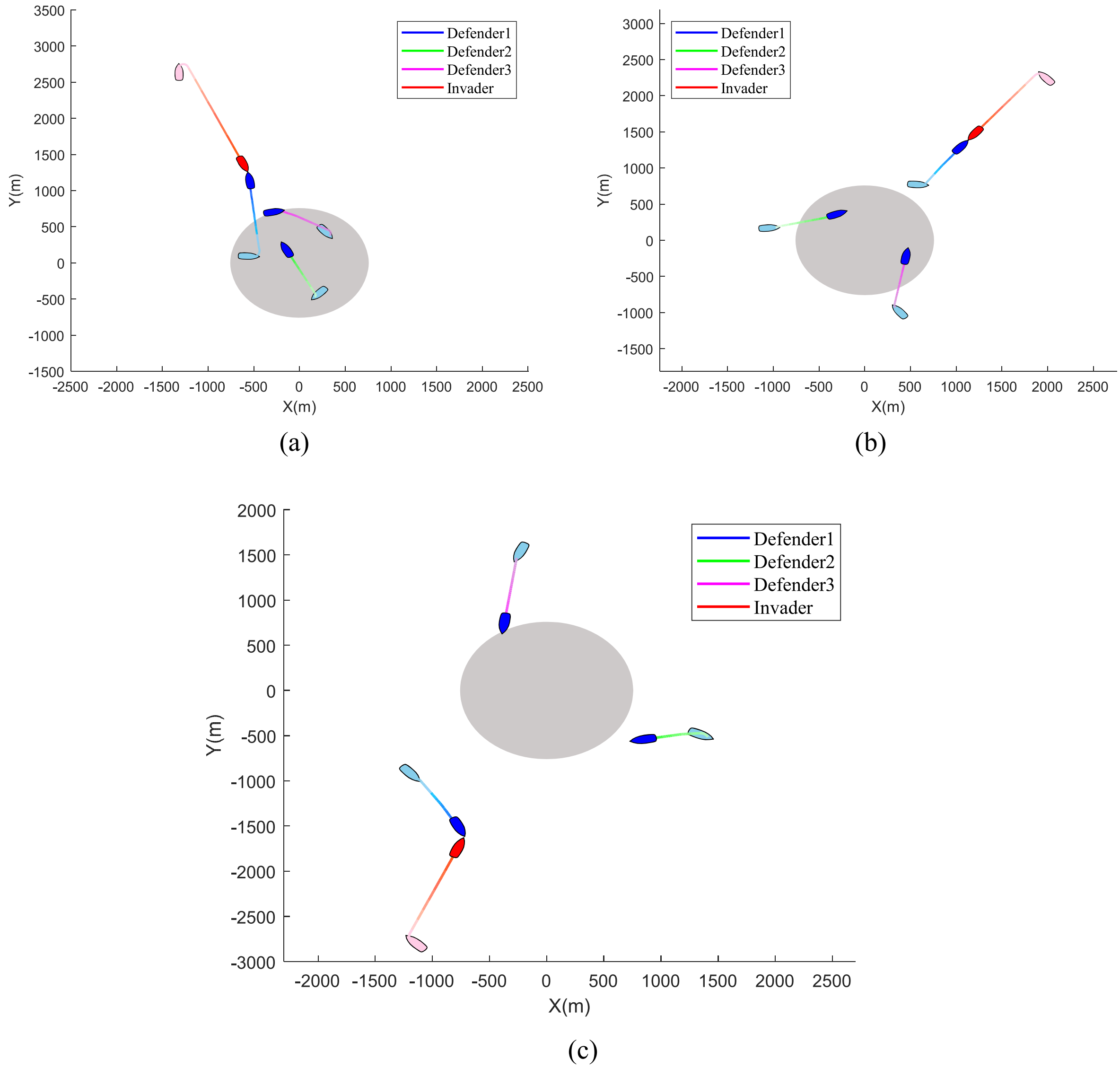


Fig. 8. Trajectories of case 1 (a) 500m (b) 1000m (c) 1500.

应对潜在威胁的回报，即通过其机动性阻止入侵者规避拦截并成功入侵。

$$r_3 = \min(\theta_d / \varphi, 1) \tag{19}$$

The comprehensive payoff function of the lower game is constructed as follows: 下层博弈的综合支付函数

$$R_l = k_4 \cdot r_4 + k_5 \cdot r_5 + k_6 \cdot r_6 \tag{20}$$

In the equation, k_4, k_5 and k_6 represent the specific gravity coefficient; r_4 represents the interception distance payoff, that is, the payoff obtained by the defender adopting the radical task by shortening the distance from the target point, as shown below:

$$r_4 = \sum_{i=1}^{n} \Delta d_i \cdot A_i \tag{21}$$

拦截距离支付

Where Δd_i represents the change of distance between the i th USV and its target point.

r_5 represents a conservative payoff, which is related to the actual proportion of the safety range in the danger range, as shown below:

$$r_5 = \min\left(\frac{\cup_{i=1}^{n}(\theta_i \cdot |A_i - 1|)}{\varphi}, 1\right) \tag{22}$$

代表保守性收益，其与安全范围在危险范围中的实际比例相关

Where θ_i represents the safety range corresponding to the current moment of the i th defender, and $\theta_i \cup \theta_j$ is the union of two safety range angles.

r_6 represents the payoff of avoiding collision, as shown below:

$$r_6 = \sum_{i=1}^{n} \sum_{j=i+1}^{n} p_{ij} \tag{23}$$

避撞的支付

Where p_{ij} represents the payoff of avoiding collision between the i th USV and the j th USV, as shown below:

$$p_{ij} = \begin{cases} 1 & , d_{ij} \geq D_2 \\ (d_{ij} - D_1) \big/ (D_2 - D_1) & , D_1 < d_{ij} < D_2 \\ 0 & , d_{ij} \leq D_1 \end{cases} \tag{24}$$

Where D_1 and D_2 are constants, and D_1 is less than D_2. d_{ij} is the distance between the i th USV and the j th USV.

4.2.4. *Solving mixed strategy Nash equilibrium*

Definition 3. Pure strategy and mixed strategy. According to whether the participant selection strategy is determined to be unique, the participant's strategy can be divided into pure strategy and mixed strategy. The pure strategy means that the participant can only select a specific action to perform at a particular moment; the mixed strategy means that

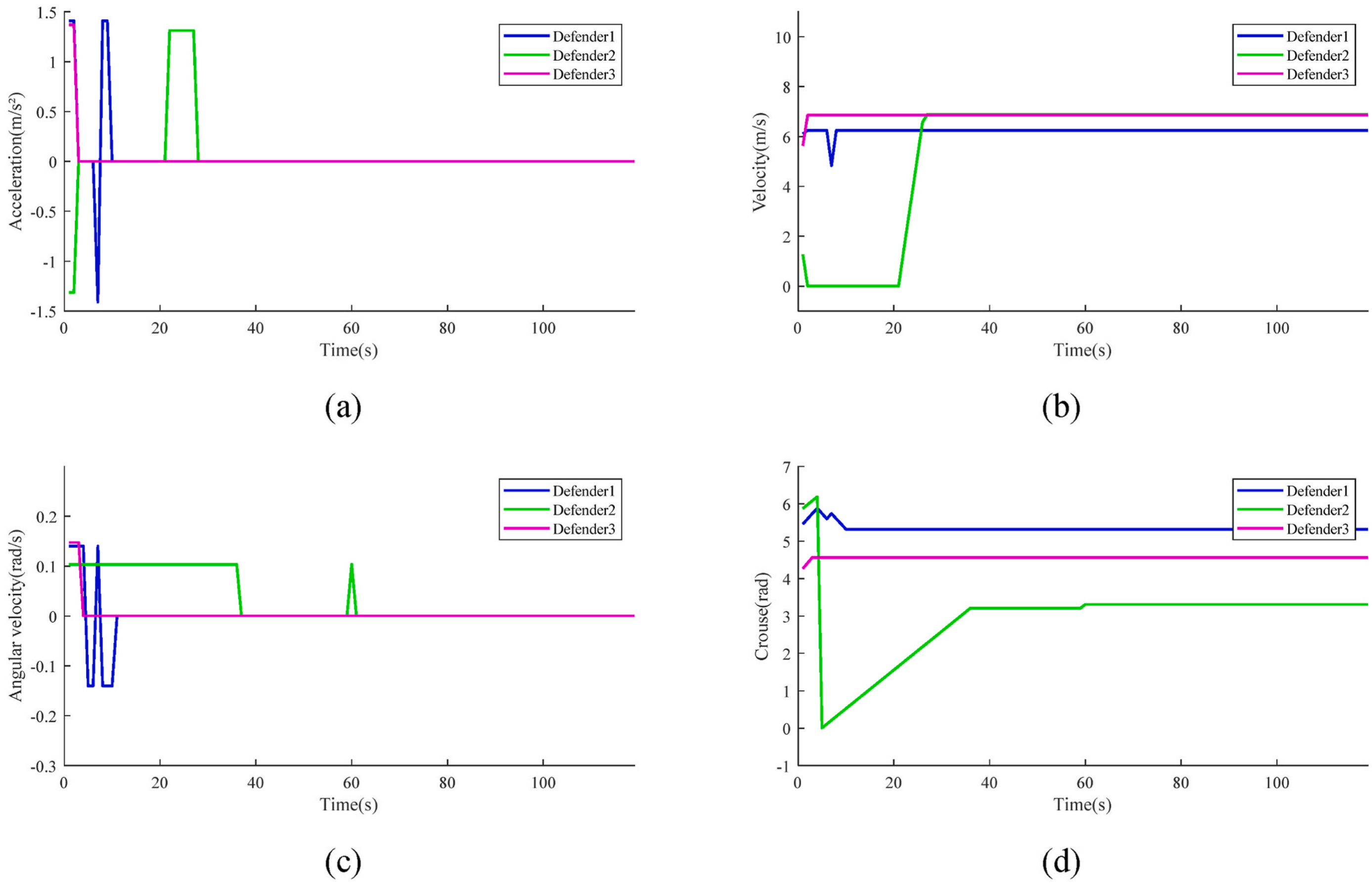


Fig. 9. The curve of motion parameters of each defender in case 1 (a) Acceleration (b) Velocity (c) Course (d) Angular velocity.

the participants randomly select an action from the set of optional actions according to the action probability, that is, the action selected each time in the same state may be different (Shao et al., 2022).

If the pure strategy Nash equilibrium is used as the strategy chosen by the participants, the behavior of the participants will be fixed. Since the payoff function relied on in the solution process is designed through factors such as interception ability, coordination ability, and potential threat, it will be affected by subjective experience. Therefore, the interception strategy obtained from the pure strategy Nash equilibrium is easily exploited and countered by the adversary. In addition, the number of pure strategies in Nash equilibrium maybe 0 or not unique (William S, 2010). In order to solve this problem, in this paper, the interception strategy of the defender is obtained by solving the mixed strategy Nash equilibrium.

Definition 4. Nash equilibrium. When all participants ' strategies reach Nash equilibrium, no participant can improve their payoffs by unilaterally changing their strategies without changing others ' strategies (Eric M, 1999).

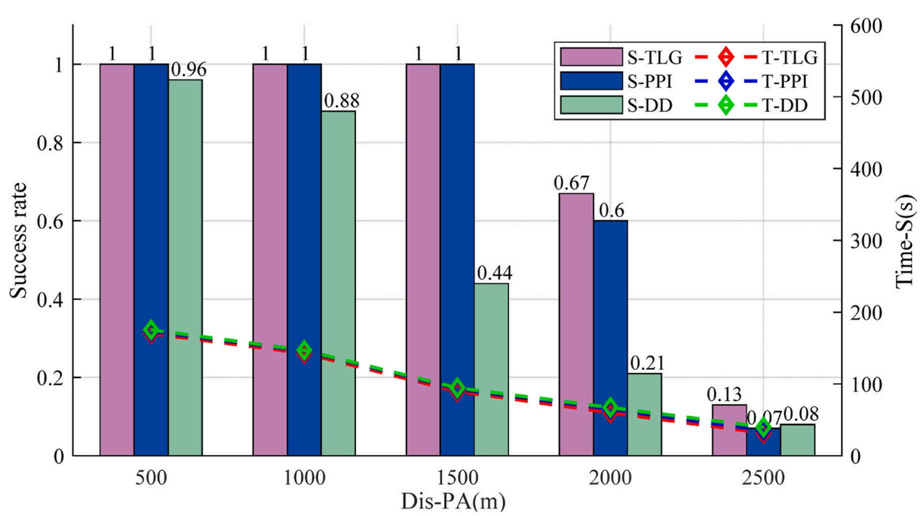


Fig. 10. Simulation results under different Dis-PA in Case1.

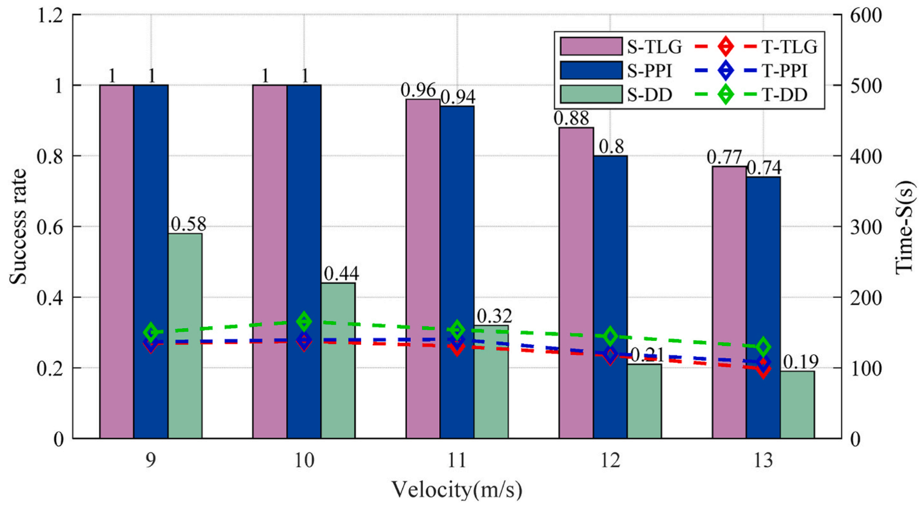


Fig. 11. Simulation results under different velocity in Case1.

Solving the game model is usually the process of solving Nash equilibrium or approximate Nash equilibrium. In this paper, the probability value of each strategy participants select is solved by linear programming, that is, the mixed strategy equilibrium solution of strategy selection. The linear programming problem is as follows (Wang and Yang, 2004):

$$\begin{aligned} &\operatorname{argmax}(z) \\ &\sum_{i}^{A_M}\sum_{j}^{A_N}\left(P_i \cdot R_{ij}\right) \geq z \\ &s.t.\begin{cases} \sum_{i}^{A_M} P_i = 1 \\ P_i \geq 0 \end{cases} \end{aligned} \tag{25}$$

In the equation, P_i represents the probability that the defenders adopts the i th strategy; R_{ij} indicates that when the defenders adopts the i th strategy and the intruder adopts the j th strategy, the payoff of the defenders. The pseudocode of the TLG algorithm is described in Algorithm 1.

Algorithm 1. TLG algorithm.

```
Input : Position, velocity, heading and maneuverability of each USV
Output : the angular velocity and acceleration of each defensive USV
for each optional strategy of the intruder in the upper game do
    Calculate the optimal interception point of each defensive USV
    for each optional strategy of the defenders in the upper game do
        Calculate the payoff under this strategy combination
    end
end
Solve the upper game model
According to the results, the optimal sub-task is selected for each defensive USV.
for each optional strategy of the intruder in the lower game do
    for each optional strategy of the defenders in the lower game do
        Calculate the payoff under this strategy combination
    end
end
Solve the lower game model
According to the results, the typical maneuver is selected for each defensive USV.
```

5. Simulation experiment

In this part, we prove the effectiveness of the TLG algorithm through Monte Carlo simulation. Monte Carlo simulation includes comparative experiments of TLG, DD (Li et al., 2022), and PPI (Meng et al., 2021). To illustrate this, we consider a scenario with a circular protected area, a defender cluster consisting of three USVs, and a target USV(invader). The invader has superior maneuverability relative to the defenders. In addition, we design two strategies for the invader, including fixed strategy and intelligent strategy, which are represented by case 1 and case 2, respectively. The intelligent strategy is designed based on the improved artificial potential field method (Huang et al., 2019).

5.1. Simulation setup

In this scenario, the defenders perform patrol tasks on a circle 1500 m away from the center of the protected area. Considering the mission requirements, the defenders are evenly distributed on the patrol path. We assume that the invader is found at a distance of 3000 m from the center of the protected area. It is believed that the center of the protected area is (0, 0). The maximum velocity of all defenders is 5m/s ~7.5m/s; the maximum angular velocity is 0.1 rad/s ~0.15 rad/s, and the capture radius is 100 m. The maximum velocity of the invader is 10m/s. The maximum angular velocity is 0.2 rad/s, and the observation radius is 1000 m. Considering the energy limitation of the USV, we set the total simulation time to 600 s. In the TLG, the maneuvering range θ is 10°, the distance parameters D_1 is 20, and D_2 is 50.

5.2. Simulations for case 1

5.2.1. Performance of TLG in case 1

Fig. 8 depicts the trajectory of the defenders intercepting the intruder when patrolling on a circular path at different distances from the center of the protected area (Dis-PA). A blue boat symbol represents the defender, and a red one represents the intruder. The light color represents the position of each USV at the beginning of the simulation, and the dark color represents the position of each USV at the end of the simulation. In addition, colors of different depths are used to describe the trajectories of each USV better to display the real-time location information of the USV. The grey area indicates the location of the protected area; Fig. 9 shows the change curve of each defender's velocity, acceleration, heading, and angular velocity when the patrol path is 1500 m away from the center of the protected area.

It can be seen from Fig. 8 that the TLG algorithm can be well applied to intercept the intruder who adopts a fixed strategy. Taking Fig. 8 (a) as

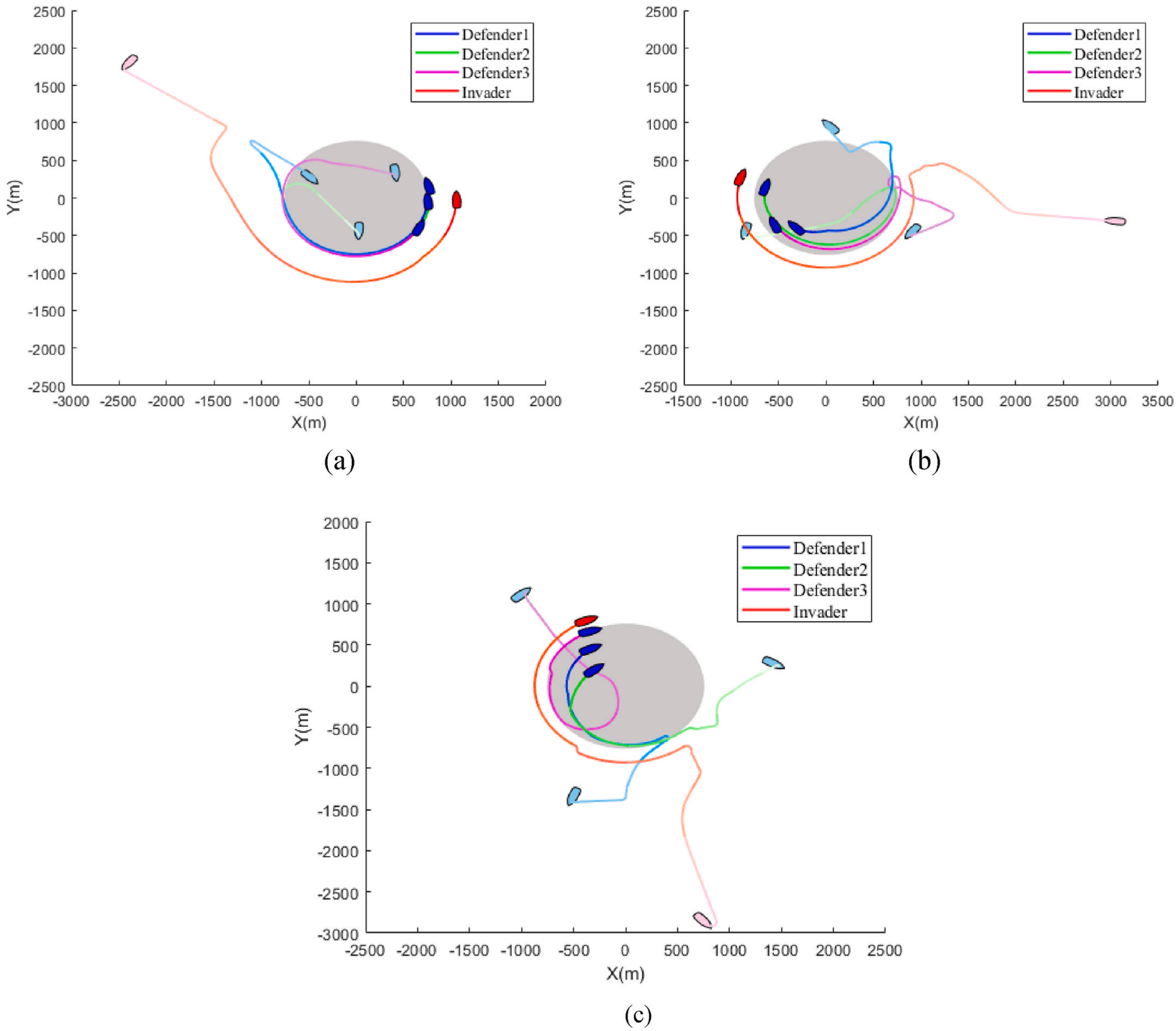


Fig. 12. Trajectories of case 2 (a) 500m (b) 1000m (c) 1500m.

an example, Defender 1 takes the radical sub-task to approach the intruder. In the 119th second, the distance between the intruder and Defender 1 is less than the capture radius, which makes the interception successful. At the same time, Defender 2 and Defender 3 employ conservative sub-tasks to deal with potential threats. In addition, Dis-PA will affect the time spent by the defender to intercept the intruder successfully (Time-S), which in turn affects the game results. Therefore, in order to explore the influence of Dis-PA on the TLG algorithm, Fig. 10 gives the game results of Dis-PA in the range of 500–2500 m.

5.2.2. *Adaptiveness of TLG in case 1*

It can be seen from Fig. 10 that when the Dis-PA is less than 1500 m, the interception success rate of TLG and PPI can reach 100 %, and the average Time-S is greater than 94.6 s and 93.1, respectively. When the Dis-PA increases to 2500 m, the interception success rate of the three algorithms decreases to about 10 %, and the average Time-S is 40 s, 35.5s, and 31.7s, respectively. The simulation results show that with the increase of the Dis-PA, the interception success rate gradually decreases, and the interception time decreases. This is because when the Dis-PA is large, it is difficult for the defenders with poor mobility to have sufficient time to complete the deployment and successfully intercept the intruder. In the round where the defenders successfully intercept the intruder, the interception time will be shorter due to the close distance between them. The DD algorithm performs poorly at each stage, while the TLG algorithm performs better. This is because both the TLG algorithm and the PPI algorithm consider the possible location of the intruder for some time in the future, while the DD algorithm does not do this. In addition, the TLG algorithm considers the angular velocity limit of each USV based on the PPI algorithm and better considers the cooperative relationship between the defenders, so it has a higher interception success rate and less Time-S.

In the process of game confrontation in the real world, there are many factors that affect the outcome of the confrontation, and it is impractical to focus only on one factor. In order to better evaluate the adaptability of the TLG, we use Monte Carlo simulation to evaluate the impact of the velocity of the invader. Fig. 11 gives the game results of the invader's velocity in the range of 9–13 m/s. As shown in Fig. 11, with the increase of the intruder speed, the interception success rate and Time-S

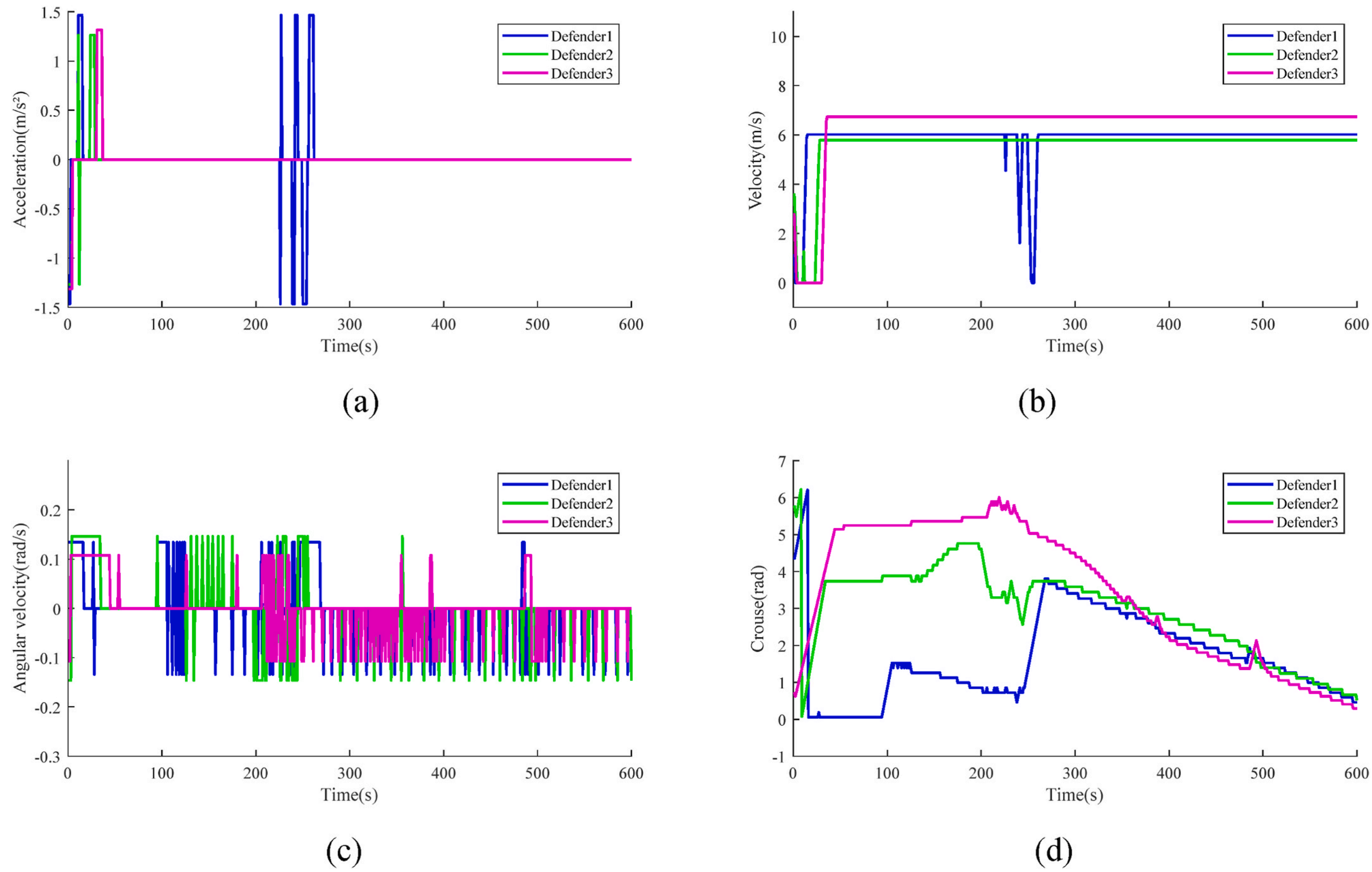


Fig. 13. The curve of motion parameters of each defender in case 2 (a) Acceleration (b) Velocity (c) Course (d) Angular velocity.

under the three algorithms are decreasing, and the Time-S corresponding to the TLG algorithm is the smallest. However, it is evident that when the TLG algorithm is used, the downward trend of the interception success rate is the least significant. Specifically, when the intruder's speed increases from 9 m/s to 13 m/s, the interception success rate of the TLG algorithm decreases by 23%, that of the PPI algorithm decreases by 26%, and that of the DD algorithm decreases by 39%. In short, the TLG algorithm has good adaptability when facing an intruder with a fixed strategy.

5.3. *Simulations for case 2*

5.3.1. *Performance of TLG in case 2*

Fig. 12 describes the trajectory of each USV corresponding to different Dis-PA when the intruder adopts an intelligent strategy. Fig. 13 depicts the change curves of velocity, acceleration, heading and angular velocity of the defenders when the Dis-PA is 1500 m.

It can be seen from Fig. 12 that under each simulation condition, each defender shows a strong cooperation trend. Taking Fig. 12 (c) as an example, at the beginning of the simulation, Defender 1 takes the radical sub-task to approach the intruder, while Defender 2 and Defender 3 perform the conservative sub-task. The simulation is carried out to the 94 th second. The intruder detects the interception behavior of Defender 1 and adjusts the heading to avoid interception. At this time, the probability that defender 1 successfully intercepts the intruder is reduced, and it turns to perform conservative sub-tasks. The simulation was carried out to the 135th second, and Defender 2 had the possibility of successful interception and changed from performing conservative sub-tasks to performing radical sub-tasks. The simulation was carried out to the 198th second. The intruder detected the interception behavior of Defender 2 and adjusted the course again to avoid interception. At this time, the possibility of successful interception of Defender 2 gradually decreased and began to perform conservative sub-tasks until the end of the simulation experiment.

5.3.2. *Adaptiveness of TLG in case 2*

More factors affect the game results when the intruder adopts an intelligent strategy. We designed several comparison conditions and evaluated them using Monte Carlo simulation. The adaptive ability of various algorithms is analyzed according to the invasion success rate and the time consumed by the round of unsuccessfully capturing the intruder (Time-UNS).

Fig. 14 (a) shows the game results of the Dis-PA in the range of 500–3000 m. When the Dis-PA is less than 1500 m, the TLG algorithm has the highest interception success rate, which can reach 100 %. When the Dis-PA is greater than 1500 m, the interception success rate of the TLG algorithm gradually decreases. However, it is still higher than the other two algorithms under the same conditions. Under various simulation conditions, the Time-UNS corresponding to the TLG algorithm is also the highest. Therefore, the TLG algorithm can have a higher interception success rate. Even in the round where the intruder is not successfully captured, the TLG algorithm can effectively prolong the time taken by the intruder to successfully enter the protected area. In general, the interception success rate of the PPI algorithm and DD algorithm increases first and then decreases with the increase of Dis-PA. In contrast, the TLG algorithm is relatively more stable. Therefore, in the face of intelligent intruders, TLG has better adaptability and robustness than the PPI and DD algorithms.

Fig. 14 (b) shows the game results of the intruder velocity in the range of 9m/s ∼ 13m/s. Fig. 14 (c) shows the game results of the maximum angular velocity of the invader in the range of 0.1 rad/s ∼ 0.5 rad/s. By analyzing Fig. 14 (b) and (c), it is found that with the increase

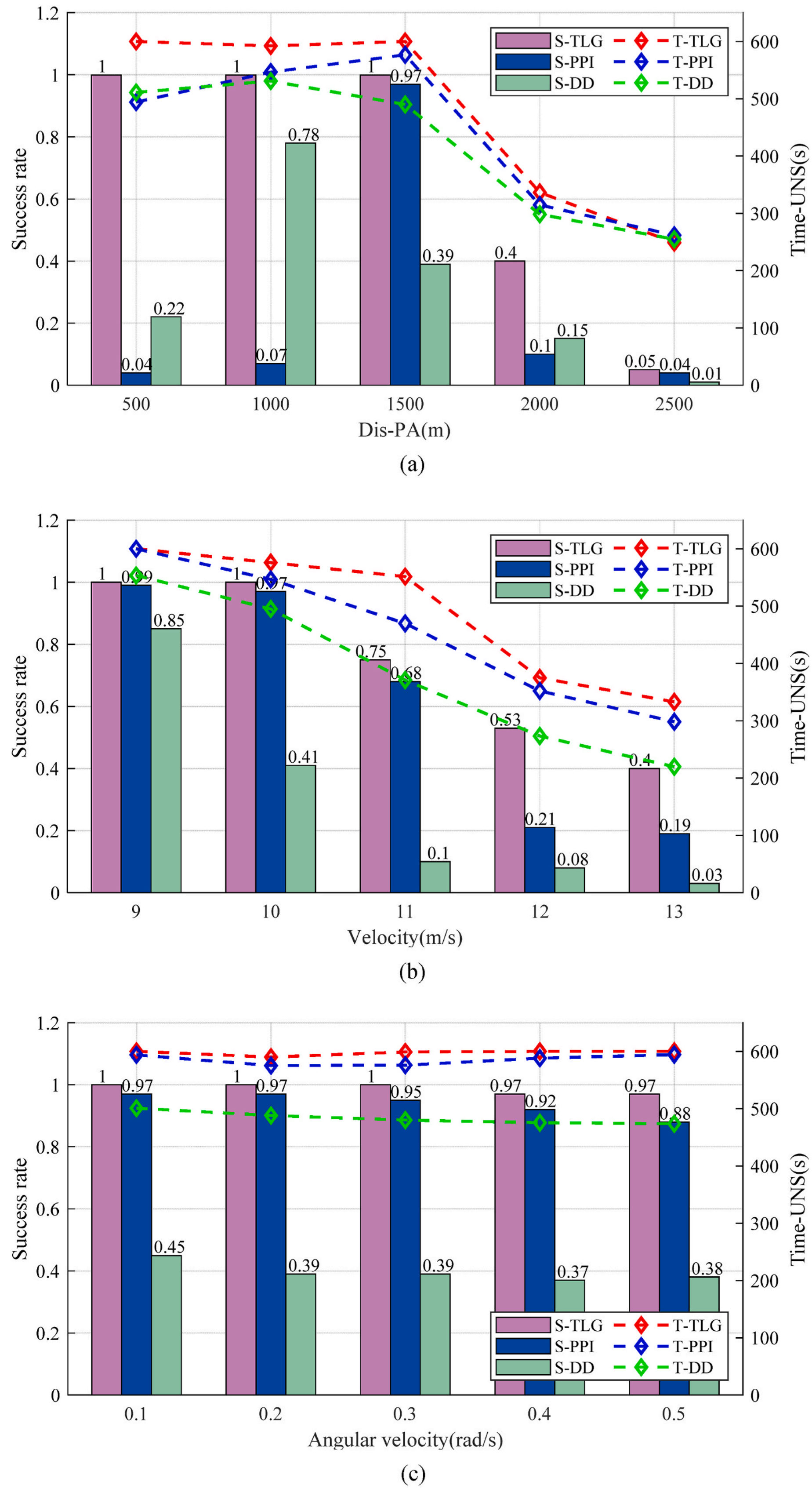


Fig. 14. Simulation results under different simulation conditions in Case2 (a) Dis-PA (b) Velocity (c) Angular velocity.

of the intruder ‘s velocity or angular velocity, the interception success rate corresponding to each algorithm decreases to varying degrees, but it is not difficult to see that the TLG algorithm has advantages under various simulation conditions. Compared with angular velocity, velocity has a more significant effect on success rate and Time-UNS.

6. Conclusion

This paper focuses on three aspects. Firstly, through the mathematical description of the reachable set of USV, the possibility of successful interception of the superior USV by each inferior defensive USV at the current moment is analyzed. Secondly, we propose two necessary conditions for completing the regional defense task when facing the superior invader under nonlinear motion. Furthermore, we simplify the complexity of the regional defense tasks by breaking them down into distinct sub-tasks aligned with the above necessary conditions. Moreover, we model the selection of sub-tasks as an upper game and the selection of basic maneuvers as a lower game to better complete the switching between different sub-tasks. Finally, Monte Carlo simulation results demonstrate the effectiveness of the proposed cooperative interception strategy for fixed and intelligent invaders.

As the number of nodes in the scene increases, the strategy combination that the TLG algorithm needs to consider will grow exponentially. This will lead to an increase in decision time, which makes the TLG algorithm unsuitable for the scene with many nodes. In addition, the research scenario of this paper is open sea area, and the influence of obstacles in the game process is not considered. Therefore, the author’s future work will focus on two aspects. On the one hand, the author will focus on achieving effective target interception in complex scenes with obstacles. Extending the single intruder interception problem to multiple intruders and further exploring the effectiveness of game theory in solving this challenge is another focus of the author’s subsequent research.

CRediT authorship contribution statement

Cong Chen: Resources, Software, Writing – original draft, Writing – review & editing. **Xiao Liang:** Conceptualization, Methodology, Project administration, Software, Supervision. **Zhao Zhang:** Conceptualization, Software. **Dianyong Liu:** Investigation, Methodology, Software. **Wei Li:** Conceptualization, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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